

Residuals from a Load-Normalized Spine-Neck Divider Reveal Limits of Scalar Isolation Metrics

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Abstract

Spine-neck resistance is meaningful only relative to dendrite-soma load. We define $\rho_L = R_{\text{neck}}/R_{\text{in},d}$, with $R_{\text{in},d}$ measured after omitting the stimulated spine, and use $\Gamma_{\text{div}} = 1/(1 + \rho_L)$ as the classical DC divider reference. Across 3,718 deterministic design-derived rows, peak local-transfer residuals were negative in 3,680/3,718 rows; pooled summaries are descriptive of the design table. Matched-time and integrated controls showed residual metric dependence. Descriptor comparisons indicated that ρ_L is a compact local-divider coordinate, whereas residuals, head amplitude, and somatic transfer require component, impedance, conductance, active-state, and downstream-filtering descriptors. The N=768 uncertainty ensemble lacked high- ρ_L rows, so a bounded high- ρ_L stress test probed that regime without population inference. Direct-matrix/DC and scoped Brian2-hosted checks supported representative implementation behavior within scope. SPINE discourages inferring electrical isolation from spine-neck geometry alone.

Keywords: dendritic spines; compartmental modeling; cable theory; voltage divider; electrical isolation; impedance; reproducible modeling

Dendritic spines place excitatory synapses in small postsynaptic compartments connected to the parent dendrite by narrow necks. These necks can impede current flow, alter local voltage, and separate spine-head depolarization from parent-dendrite or somatic impact [1–9]. A neck resistance by itself, however, is incomplete: the same series resistance has different consequences depending on the load presented by the dendritic and somatic circuit.

The first-order electrical expectation is the classical voltage divider. If a spine-head voltage drives a series neck resistance R_{neck} into a dendrite-soma load with input resistance $R_{\text{in},d}$, the DC small-signal local transfer is

$$\Gamma_{\text{div}} = \frac{R_{\text{in},d}}{R_{\text{neck}} + R_{\text{in},d}} = \frac{1}{1 + R_{\text{neck}}/R_{\text{in},d}}. \quad (1)$$

For compact notation, we denote the load-normalized spine-neck ratio $R_{\text{neck}}/R_{\text{in},d}$ as ρ_L . The abbreviation SMI is retained in some code paths, tables, and provenance files as author-defined shorthand for the same ratio, not as a claim that the ratio is a field-standard morphology index or a new cable-theory law.

This distinction matters because the divider expectation is not the same as a transient synaptic peak response. Conductance-based excitation changes driving force while the response evolves; ca-

capitance, local impedance, dendrite-to-soma filtering, active conductances, morphology, and measurement uncertainty can all alter observed voltages [10–14]. A scalar ratio can be useful when the target is local low-frequency isolation, yet fail for head amplitude, somatic voltage, nonlinear response, or categorical class assignment.

The present study therefore reframes SPINE around the residual from the analytic divider:

$$\Delta_{\text{div}} = \Gamma_{h \rightarrow d} - \Gamma_{\text{div}}. \quad (2)$$

The ratio defines the expected local divider coordinate; the residual tests where the full simulated response leaves that coordinate. This treats departures from the divider as domain-of-validity information rather than failed confirmation. Large residuals, equal-ratio nonequivalence, and amplitude failures identify the electrical variables that the compact ratio cannot contain.

We ask five bounded questions. First, do the workflow-specified low-, intermediate-, and high-isolation reference cases reproduce the intended passive responses? Second, how far do observed peak local transfers depart from Γ_{div} across existing passive, active, uncertainty, and exploratory rows? Third, when is the ratio more useful than its component resistances, and when do component, impedance, dynamic, or conductance descriptors carry additional information? Fourth, how should deterministic uncertainty and radius-error screens be reported without implying population inference? Fifth, do independent matrix and analytic benchmarks support the computational credibility of the from-scratch passive implementation? The resulting thesis is deliberately scoped: the load-normalized ratio recovers the classical local divider expectation, and SPINE maps the residual domains where a single scalar is insufficient.

The practical implication is that experimental and modeling reports should not treat spine-neck geometry alone as evidence for electrical isolation. They should report neck resistance together with dendrite-soma load, use Γ_{div} as a transparent first-order reference, prefer continuous ratio values when radius uncertainty makes threshold classes fragile, and add timing, impedance, conductance, and downstream-load descriptors before interpreting head amplitude or somatic impact.

1 Results

1.1 Reference cases reproduce the intended passive ordering

The baseline reference configuration generated the intended progression from near-isopotential head-dendrite responses to strong local isolation (Figures 1 and 2). The simulations used the baseline parameters in Table 1, $\Delta t = 0.01$ ms, a single head synapse at 20 ms, and the analysis-defined 50 ms metric window. All nine low-, intermediate-, and high-isolation target values passed the workflow-specified tolerance rule.

The low- ρ_L case produced $\rho_L = 0.00881$, $\Gamma_{h \rightarrow d} = 0.984$, $\Gamma_{h \rightarrow s} = 0.357$, and $A_h = 1.82$ mV. The intermediate case produced $\rho_L = 0.115$, $\Gamma_{h \rightarrow d} = 0.751$, $\Gamma_{h \rightarrow s} = 0.273$, and $A_h = 2.36$ mV. The high- ρ_L case produced $\rho_L = 1.32$, $\Gamma_{h \rightarrow d} = 0.102$, $\Gamma_{h \rightarrow s} = 0.0376$, and $A_h = 15.1$ mV. These target cases support implementation consistency for the reference equations, units, synaptic normalization, solver, and metric definitions in the compact passive architecture; they do not calibrate the model to a specific biological neuron.

The same traces show why the divider residual is needed. The baseline divider predictions were 0.991, 0.897, and 0.431 for the low, intermediate, and high cases. The corresponding residuals $\Gamma_{h \rightarrow d} - \Gamma_{\text{div}}$ were -0.00729, -0.1458, and -0.3286. The low case was close to the DC divider, whereas the intermediate and high transient peak ratios fell well below it.

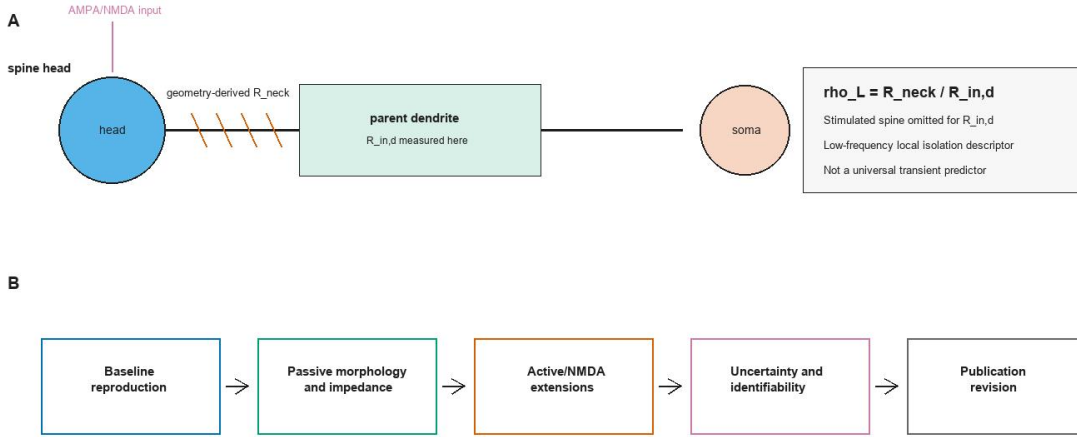


Figure 1: SPINE reference architecture and residual workflow. A conductance-based excitatory input drives an explicit spine head coupled to the parent dendrite through a geometry-derived neck resistance. The dendrite-soma input resistance is measured after omitting the stimulated spine, yielding $\rho_L = R_{neck} / R_{in,d}$ and the analytic local divider expectation $\Gamma_{div} = 1 / (1 + \rho_L)$. SMI is retained only as legacy shorthand in code and provenance files.

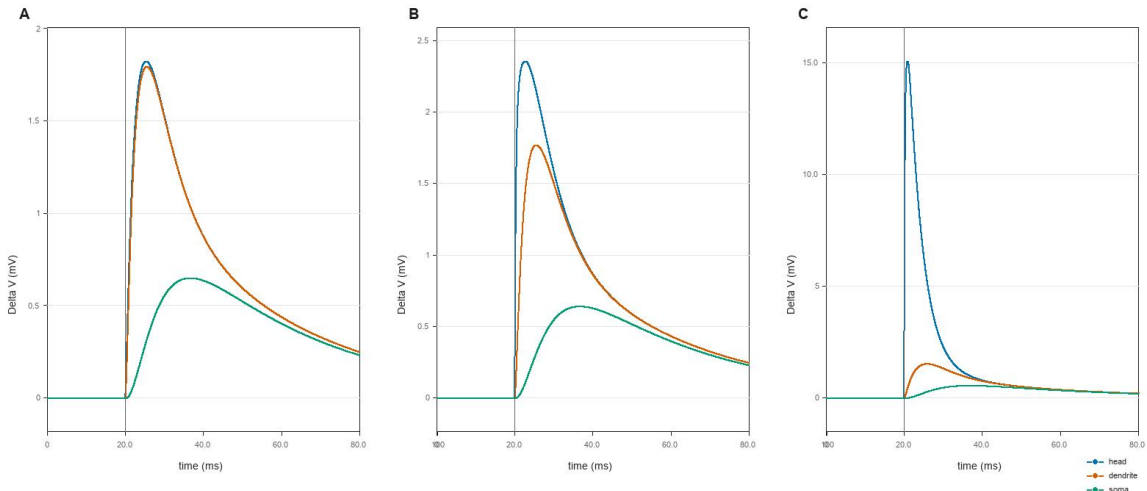


Figure 2: Baseline passive reference traces. Low-, intermediate-, and high- ρ_L traces show the intended local coupling progression. The reference cases also demonstrate that target-case ordering does not imply exact equality to the DC divider: residuals are small for the low case and larger for intermediate/high transient peak responses.

1.2 The analytic divider is a reference, not the peak-response result

Across 3,718 deterministic design-derived residual rows, the median absolute residual from the divider was 0.0548, the mean absolute residual was 0.0734, and the root-mean-square residual was 0.103. The maximum absolute residual was 0.492. Residuals were negative in 3,680/3,718 rows and positive in 38 rows, so the DC divider overpredicted this separate-peak conductance-based transfer metric in most designed rows (Figure 3). Pooled values describe this deterministic design table rather than a biological population.

The largest departures were consistent with the intended distinction between a DC reference and a transient conductance-driven peak response. The matched-neck heterogeneous-load sweep produced the largest absolute separate-peak residual, with $\rho_L = 0.452$, observed $\Gamma_{h \rightarrow d} = 0.196$, divider prediction 0.689, and residual -0.492. Fixed-load high- ρ_L rows also showed large relative departures.

Matched-time and integrated controls showed that residual magnitude is metric-dependent. In the three baseline reference cases, area-under-curve residuals were closer to the divider for intermediate/high cases than the primary separate-peak residual, whereas the matched-head-peak control emphasized timing and downstream filtering. The largest matched-neck heterogeneous-load case showed the same metric dependence: the separate-peak residual was -0.492, the matched-head-peak residual was -0.637, and the AUC residual was -0.101. Thus the divider overprediction statement applies to this separate-peak conductance-response metric, not to every possible transfer summary.

Excluding exploratory rows left the same conclusion: 3584 non-exploratory rows had median absolute residual 0.0563, mean absolute residual 0.0748, and only two positive residual rows. The divider relation therefore remains the correct analytic baseline, but the residual field is a design-specific simulation-domain map.

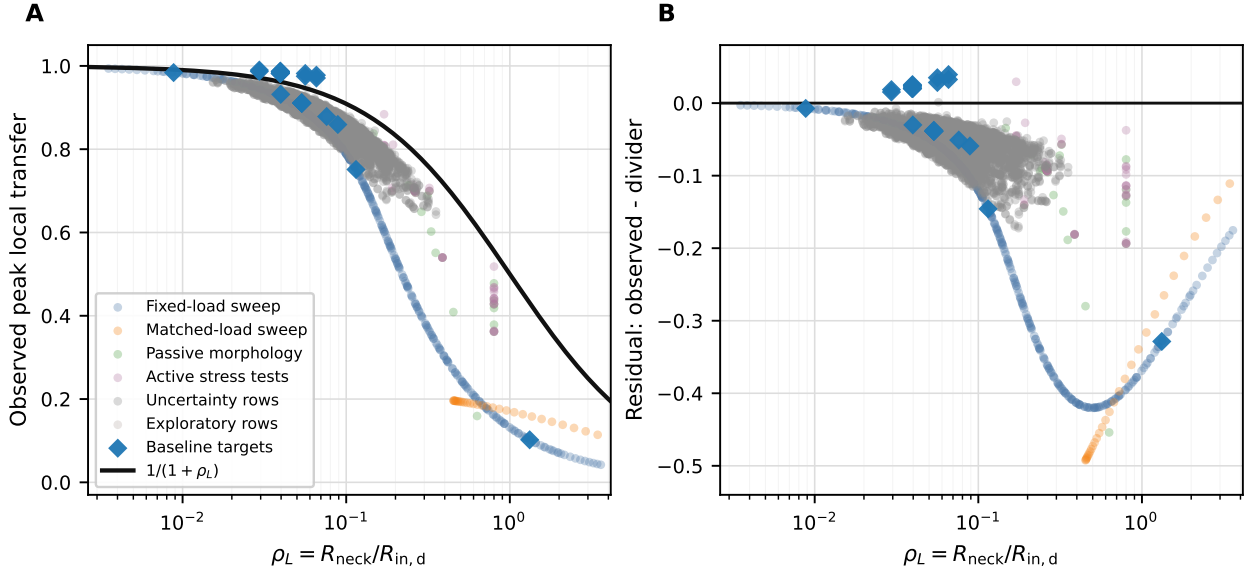


Figure 3: Analytic divider overlay and residual domain. Observed peak local transfer is compared with the DC divider expectation $\Gamma_{\text{div}} = 1/(1 + \rho_L)$ across 3,718 deterministic design-derived rows. Most points fall below the divider curve for the separate-peak metric, and the residual panel makes the design-specific domain visible. Baseline targets are highlighted; exploratory rows are included only as separated stress-test context.

1.3 Load normalization is useful but compact

The fixed-load geometry sweep behaved as a constructional sanity check. Longer and narrower necks increased R_{neck} , increased head depolarization, and decreased downstream transfer. Because $R_{\text{in},d}$ was fixed, ρ_L and R_{neck} had identical rank ordering. This result should not be read as evidence that the ratio adds information under fixed-load conditions.

The matched-neck experiment held $R_{\text{neck}} = 100 \text{ M}\Omega$ while varying dendritic leak from 0.1 to 30 nS (Figure 4). Under this manipulation, neck resistance alone could not rank the cases because there was only one neck-resistance value. Changing the dendritic load changed $R_{\text{in},d}$, and therefore changed ρ_L and downstream transfer. The load term is therefore mechanistically meaningful, but only within the target it was designed to represent.

Descriptor comparisons made this compactness explicit. For non-exploratory local transfer, the load-normalized ratio and Γ_{div} had the same absolute Spearman association because the divider is a monotone transform of the ratio, but the analytic divider was the better direct scalar prediction (cross-validated RMSE 0.052 versus 0.091 for raw ρ_L). The log component pair improved on raw ρ_L but did not outperform the analytic divider for first-order local transfer. Thus the ratio is useful as the coordinate defining the divider, not as a universally superior scalar.

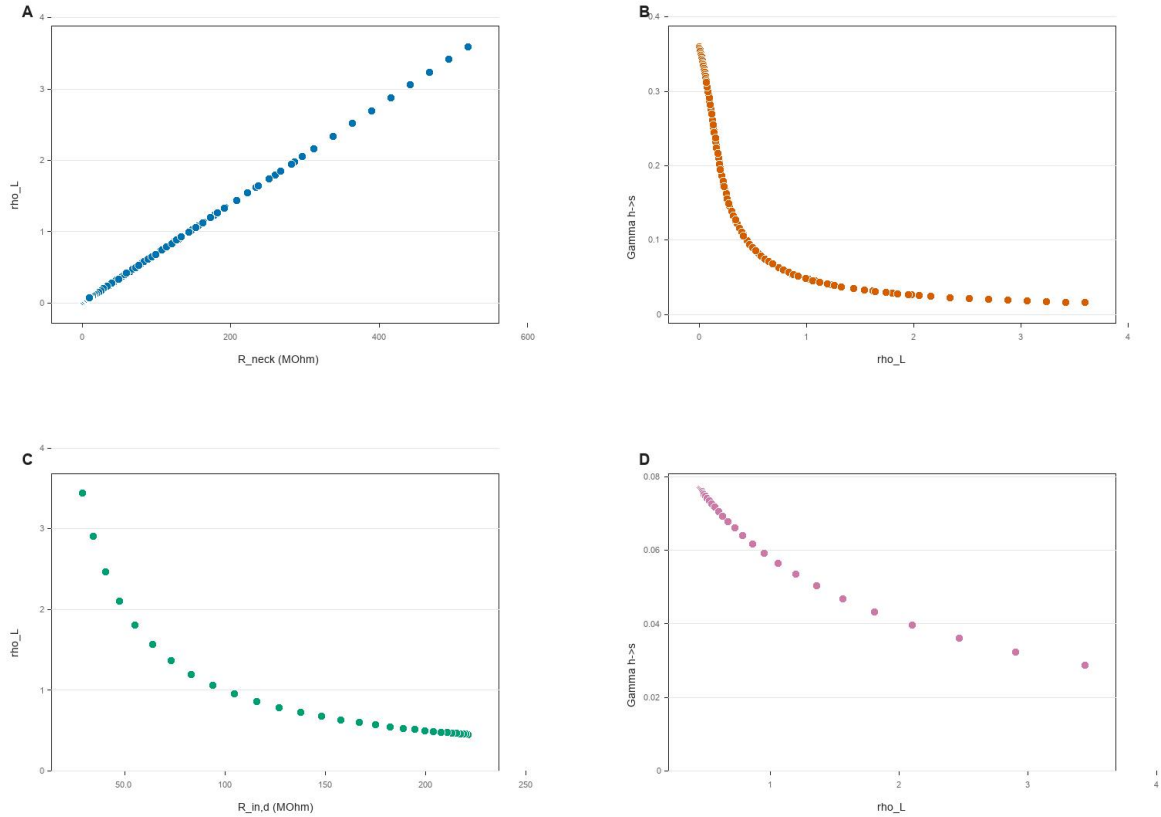


Figure 4: Fixed-load and matched-neck controls. At fixed dendritic input resistance, ρ_L is a scaled neck-resistance coordinate. When R_{neck} is held fixed and dendritic load changes, the ratio changes with $R_{\text{in},d}$ and separates cases that neck resistance alone cannot order.

External summary anchors as a bounded load-normalization decision gate. A Codex-only external-data gate recovered summary-level external data, not paired per-spine Harnett measured rows. Three summary-usable rows were eligible: a Harnett model/summary high-isolation anchor, a Popovic measured summary anchor, and a Popovic model-fit reference [6, 13]. Source-native calculations placed the Harnett anchor at $\rho_L = 4$ and $\Gamma_{\text{div}} = 0.200$ (high), the Popovic measured summary at $\rho_L = 0.0982$ and $\Gamma_{\text{div}} = 0.911$ (low), and the Popovic model-fit row at $\rho_L = 0.0718$ and $\Gamma_{\text{div}} = 0.933$ (low), so the anchors occupied distinct source-native ρ_L regimes. The separation persisted under fixed/recovered common loads: Harnett remained high, Popovic measured remained low, and Popovic model-fit remained low or intermediate. The Popovic measured uncertainty envelope remained low. Transfer ratios were treated only as compatibility proxies and not SPINE residuals. These results support a summary-level load-normalization hypothesis for part of the apparent high- versus low-isolation contrast, but they are not a per-spine reanalysis, not biological validation, not a prevalence estimate, and not a controversy resolution (Supplementary Figure S10).

1.4 Residuals, amplitude, and somatic transfer require richer descriptors

The residual analysis separated local-divider usefulness from broader prediction. For signed and absolute divider residuals, the dynamic ratio descriptor recorded as `dynamic_SMI_abs` in code/provenance tables was the strongest available scalar by rank association (absolute Spearman 0.853), while R_{neck} exceeded raw ρ_L (about 0.765 versus 0.704). In rows with synaptic conductance scale, conductance plus log-component models sharply reduced residual prediction error. Residuals are therefore better treated as a dynamic/conductance/impedance problem than as evidence that the DC ratio alone is sufficient.

Head amplitude depended on a different descriptor family. Across non-exploratory rows, synaptic conductance scale had absolute Spearman association 0.982 with A_h , whereas raw ρ_L was only 0.143. Somatic transfer depended on downstream filtering: transfer gain and transfer impedance exceeded raw ρ_L for $\Gamma_{h \rightarrow s}$ in the descriptor summary. These results support a target-specific descriptor interpretation: local divider reasoning, head amplitude, residuals, and somatic transfer require different descriptor families.

Designed passive counterexamples sharpened this interpretation. Passive equal- ρ_L cases with different morphology produced relative spread 0.28 in $\Gamma_{h \rightarrow d}$ and 0.61 in A_h , both exceeding the analysis-defined 20% failure threshold. Equal ρ_L is therefore not electrical equivalence, because the same ratio can hide different absolute resistances, capacitances, impedance spectra, synaptic drives, and downstream morphologies.

1.5 Active and NMDA stress tests preserve scope limits

Active-mechanism verification passed all 12 analysis-defined rows, including AMPA peak normalization, NMDA magnesium-block monotonicity, NMDA current-voltage relief, active-current signs, bounded gates, resting-state stability, timestep refinement, limiting-case solver comparison, strong-drive stability, and unit conversion checks. Active mechanisms were disabled in the reference baseline and enabled only in active-extension protocols.

In the designed active nonlinear suite, ρ_L remained strongly associated with local $\Gamma_{h \rightarrow d}$, but active mechanisms did not make the static ratio a general amplitude or somatic-transfer predictor (Figure 6). For active $\Gamma_{h \rightarrow s}$, input impedance at 50 Hz exceeded raw ρ_L (0.764 versus 0.339). For active A_h , frozen-gate transfer gain exceeded raw ρ_L (0.736 versus 0.236). AMPA+NMDA protocols with full restrained active conductances produced equal- ρ_L relative spread 0.86 in A_h and 0.35 in

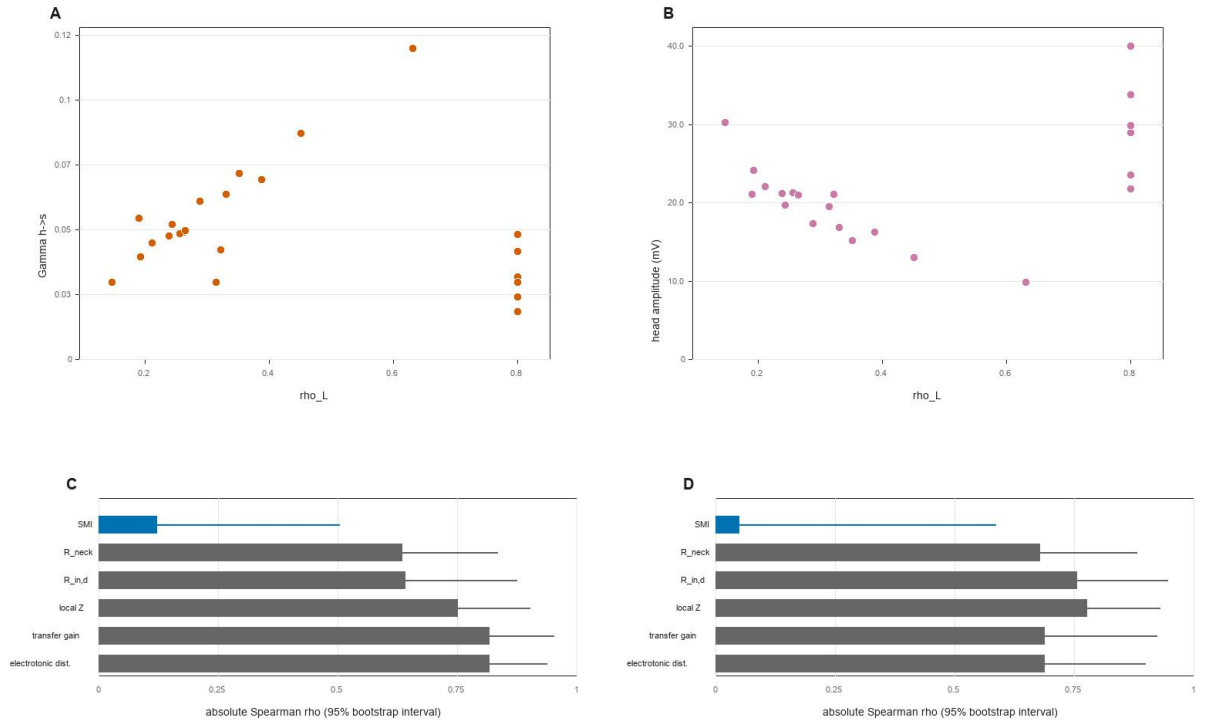


Figure 5: Passive limits of the compact ratio. Passive challenge rows show that local-divider information does not automatically transfer to head amplitude or somatic voltage. Component, impedance, transfer-gain, and synaptic-drive descriptors retain information that the ratio discards.

$\Gamma_{h \rightarrow d}$, exceeding the analysis-defined active stress-test thresholds.

Thus active mechanisms do not overturn the divider framing; they make its scope more important. Voltage-dependent current and active state add response variables that are not contained in a static morphology/load ratio.

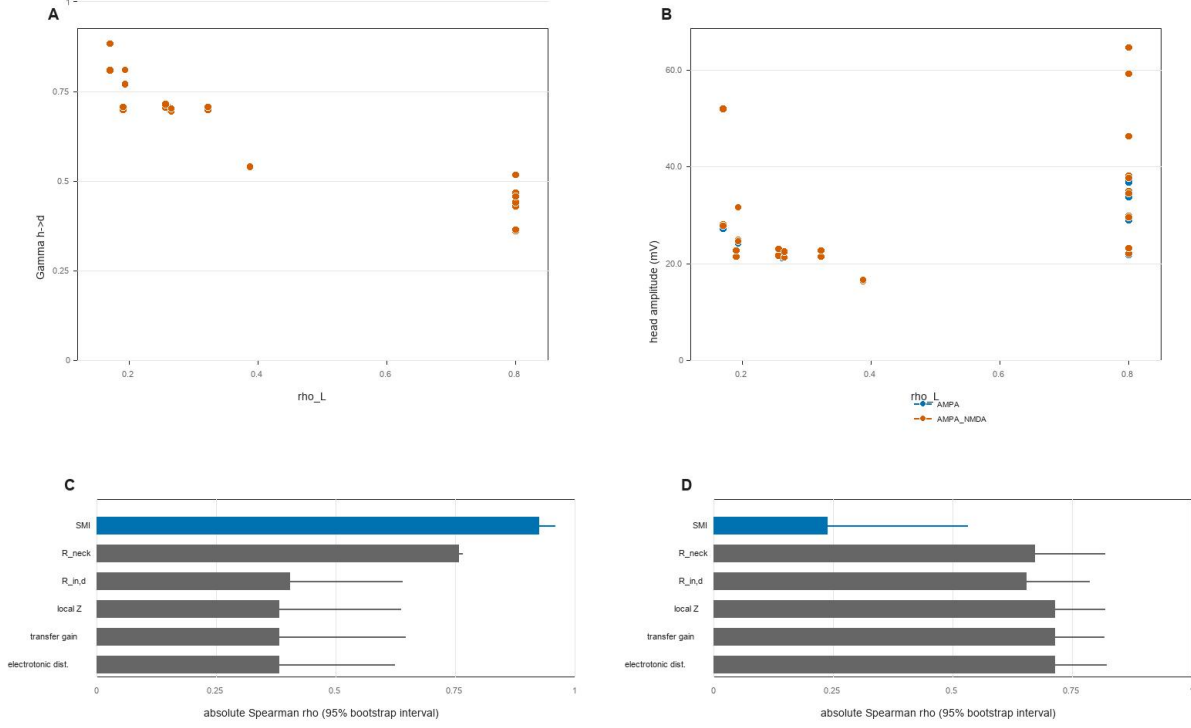


Figure 6: Active and NMDA stress-test boundaries. Active and NMDA-enabled protocols retain local ratio ordering in designed cases but increase nonequivalence for amplitude and other response targets. These rows are generic stress tests rather than cell-type-calibrated active reconstructions.

1.6 Deterministic uncertainty stabilizes sampled low/intermediate behavior but not high-ratio claims

Progressive uncertainty analysis resolved the small-sample instability seen in the original 48-versus-96 comparison. By $N=768$, final ρ_L median convergence passed: the 384-to-768 relative change was 0.012, below the analysis-defined 0.10 threshold. The maximum final median relative change across tracked outputs was 0.029. Exact top-predictor ranking remained less stable than descriptor-family interpretation.

The $N=768$ ensemble does not support high- ρ_L uncertainty claims. It contained 757 low- ρ_L rows, 11 intermediate- ρ_L rows, and zero high- ρ_L rows. A bounded high- ρ_L extension added in the final revision sampled 41 true high-ratio parameter combinations and 205 deterministic radius-measurement rows; 16 crossed below $\rho_L = 0.75$, giving a design flip fraction of 0.078, and the median peak residual remained negative at -0.3245. This extension samples the missing regime as a deterministic stress test, but it is not a replacement for a full uncertainty ensemble. High-isolation conclusions therefore rest on baseline reference, designed challenge rows, and explicitly

labeled targeted checks.

Radius-only uncertainty produced 184 class flips out of 768 sampled parameter combinations, a design fraction of 24.0%. Low-class combinations flipped in 22.9% of cases, all 11 intermediate combinations flipped, and no high- ρ_L combinations were present. Because R_{neck} scales as $1/r_n^2$, continuous ratio values with explicit radius uncertainty are safer than treating low, intermediate, and high labels as sharply separated biological states.

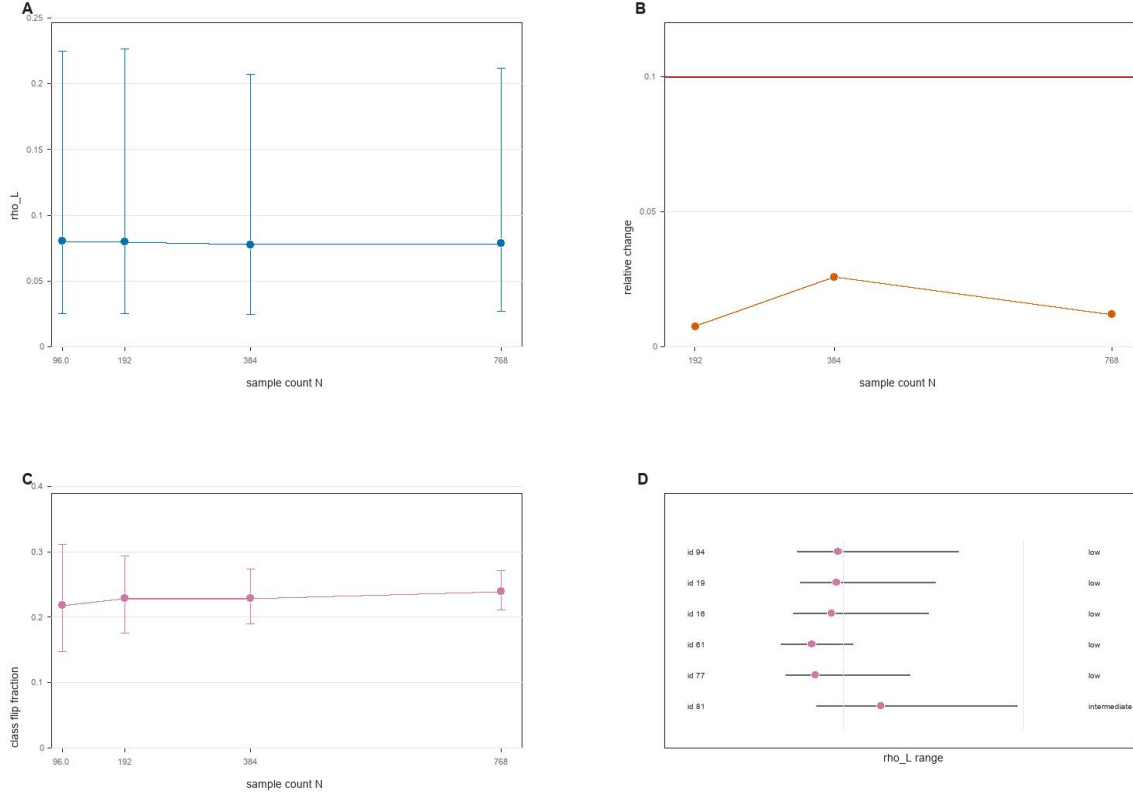


Figure 7: Deterministic uncertainty and radius sensitivity. Progressive sampling stabilized continuous low/intermediate ρ_L summaries by $N=768$, while radius-error propagation exposed threshold-class fragility. Fractions are design fractions over sampled parameter combinations, not biological rates.

1.7 Independent benchmarks support computational credibility within scope

Computational credibility was assessed beyond internal scheme agreement. An independently written direct-matrix implementation of the same passive three-compartment baseline reproduced existing SPINE baseline traces to numerical round-off: the maximum all-trace absolute voltage difference was 1.25×10^{-13} mV, the maximum head-amplitude difference was 1.42×10^{-14} mV, and the maximum $\Gamma_{h \rightarrow d}$ difference was 7.55×10^{-15} . DC analytic checks reproduced the two-node spine-omitted $R_{\text{in},d}$ value of 144.486692 M Ω to round-off and confirmed the divider limits.

Backward Euler versus Crank-Nicolson peak differences were small in the baseline reference cases: maximum peak differences were 0.00203 mV for A_h , 0.000534 mV for A_d , 0.000189 mV for A_s , and

0.000500 for $\Gamma_{h \rightarrow d}$. These are internal numerical self-consistency checks. Brian2 was installed as an optional project-local validation dependency and used for Brian2-hosted scoped checks of the three baseline cases, one distributed-neck passive case, and one AMPA/NMDA synapse-only case; all compared metrics passed predefined tolerances. Because the successful path used Brian2 for the validation environment and state hosting rather than native Brian2 integration of the full SPINE model, these checks are not all-row or active-channel cross-simulator validation. NEURON remains unavailable and is not claimed.

The final package includes manifest and checksum workflows and points to the verified public release at <https://github.com/blakepi/spine-divider-residuals>, release tag **v1.0.0-submission**, and archival DOI <https://doi.org/10.5281/zenodo.20672333>.

Table 1: Baseline reference passive parameters. Values are taken from the manuscript-faithful baseline configuration and the three analysis-defined reference neck geometries. Parameters are idealized passive-reference values chosen to span low, intermediate, and high load-normalized ratio regimes, not cell-type-calibrated biological estimates.

Parameter	Meaning	Value
E_L	leak reversal	-70 mV
E_{syn}	excitatory reversal	0 mV
c_m	specific capacitance	0.01 pF μm^{-2}
$\bar{g}_L$	head specific leak	3×10^{-6} nS μm^{-2}
r_h	head radius	0.35 μm
C_d, C_s	dendrite and soma capacitance	120 pF, 220 pF
$g_{L,d}, g_{L,s}$	dendrite and soma leak	2.5 nS, 7.0 nS
g_{ds}	dendrite-soma coupling	12.0 nS
ρ_i	intracellular resistivity	100 Ω cm
g_{max}	peak synaptic conductance	1.4 nS
τ_r, τ_d	rise and decay constants	0.3 ms, 3.0 ms
$t_0, \Delta t$	event time and time step	20 ms, 0.01 ms
T	metric window	50 ms
$R_{\text{in},d}$	spine-omitted dendrite-soma input resistance	144.487 M Ω
low neck	$L_n, r_n, R_{\text{neck}}$	0.25 μm , 0.25 μm , 1.273 M Ω
intermediate neck	$L_n, r_n, R_{\text{neck}}$	0.75 μm , 0.12 μm , 16.579 M Ω
high neck	$L_n, r_n, R_{\text{neck}}$	1.50 μm , 0.05 μm , 190.986 M Ω

2 Discussion

SPINE now supports a residual-domain interpretation of the load-normalized spine-neck ratio. The ratio $\rho_L = R_{\text{neck}}/R_{\text{in},d}$ encodes the classical local divider expectation $\Gamma_{\text{div}} = 1/(1+\rho_L)$; SMI remains only as legacy shorthand for this quantity in code-facing outputs. The scientific work begins after that analytic reference is stated: transient conductance-based simulations usually produced peak local transfer below the DC divider, and the residuals identify where a single scalar loses information.

This reframing is important because it separates a true but limited electrical expectation from broader prediction. The low- ρ_L reference case was close to the divider limit, while intermediate and high reference cases were substantially below the divider in peak transfer. Across the full derived table, residuals were usually negative and sometimes large for the separate-peak metric. That pattern is not a failure to reproduce the divider; it is evidence that the divider is the low-frequency baseline and that peak synaptic responses live in a richer dynamic system.

The practical use pattern is therefore two-step. Use the load-normalized ratio to report first-pass local head-to-dendrite divider context. Then report residuals, component resistances, impedance, transfer gain, synaptic conductance scale, active state, morphology, and uncertainty whenever the target is peak transfer, head amplitude, somatic voltage, nonlinear response, class membership, or electrical equivalence. For experimental interpretation, that means reporting neck resistance together with dendritic/somatic load rather than geometry alone, and treating Γ_{div} as a transparent reference rather than as evidence that an observed transient response has been explained.

The ratio succeeds locally because it compares a series neck resistance with the load driven by that resistance. This is the classical cable-theoretic intuition for spine-neck isolation [1, 2, 4, 10]. It also explains why the matched-neck control is meaningful: when R_{neck} is fixed, different dendritic loads produce different divider expectations.

The same logic explains why the ratio fails outside its target. Head amplitude depends on absolute synaptic conductance, driving-force reduction, local capacitance, and input impedance. Somatic transfer adds dendrite-to-soma filtering, a distinction related to location-dependent dendritic propagation and somatic EPSP normalization [15]. Active state, NMDA voltage dependence, clustered input, and timing can change voltage trajectories without changing the morphology-derived ratio. Equal ρ_L cannot imply electrical equivalence because many circuits share a ratio while differing in absolute RC constants, impedance phase, conductance drive, and morphology.

This does not make the ratio useless. It makes it conditional. A scalar descriptor can be valuable when its target is explicit and its residuals are measured. It becomes misleading only when the local divider coordinate is stretched into an amplitude, somatic, nonlinear, or categorical claim.

The radius audit highlights a measurement boundary. Because $R_{\text{neck}} \propto 1/r_n^2$, modest radius error can produce large resistance and ratio error, and spine-neck geometry is itself plastic and difficult to measure precisely [16, 17]. The N=768 radius screen produced a 24.0% design-fraction class flip, with all 11 intermediate- ρ_L combinations crossing a class boundary. This supports continuous reporting with explicit uncertainty more strongly than thresholded low/intermediate/high labels.

The same N=768 ensemble contained no high- ρ_L rows. This absence should be treated as a design limitation. The targeted high- ρ_L extension added here is useful as a bounded stress test, but it is not a substitute for a full uncertainty ensemble spanning that regime. High-isolation behavior is therefore supported by reference cases, designed challenge rows, and targeted checks, whereas uncertainty stability is supported only for sampled low/intermediate combinations.

The from-scratch passive implementation is computationally credible within the scope tested. Internal unit, current-balance, limiting-case, timestep, impedance, and BE-versus-CN checks support numerical self-consistency. More importantly for external credibility, an independent direct-matrix implementation reproduced baseline passive traces and metrics to round-off, and DC analytic benchmarks checked the dendrite-soma load and divider limits.

Brian2 added a useful optional implementation-check anchor, but the scope remains narrow. Brian2 was installed in an isolated validation environment and the Brian2-hosted scoped checks reproduced representative passive baselines, one distributed-neck passive case, and one AMPA/NMDA synapse-only case within predefined tolerances. However, native Brian2 state updaters were not the final successful integration path for the stiff compact circuit, and restrained active channels were not reproduced in Brian2. The strongest supported statement is therefore narrower than global simulator validation: the passive reference equations are consistent with independent matrix/DC benchmarks and representative Brian2-hosted checks. Extending native cross-simulator checks to broader morphologies, active protocols, and uncertainty rows remains useful future work.

The results fit naturally with mixed experimental evidence about spine electrical isolation. Some optical and uncaging studies support substantial compartmentalization in particular contexts, whereas other work emphasizes modest isolation, indicator limits, and context-dependent interpre-

tation [5, 7, 9, 13, 14]. A manuscript that reports only a single ratio would flatten these differences. A manuscript that reports the ratio, divider expectation, residuals, and target-specific descriptors can connect morphology to physiology without overclaiming.

The summary-level external decision gate reinforces the reporting lesson without broadening the validation claim. The machine-recoverable Harnett and Popovic anchors suggest that some apparent high- versus low-isolation contrast can be reframed in load-normalized coordinates, but the evidence remains summary-level only: Harnett is a model/summary anchor rather than recovered paired measured per-spine data, Popovic measured and model-fit rows are kept separate, Kwon and Cornejo remain context/exclusion rows, and transfer proxies are not SPINE residuals. The overlay motivates reporting dendritic load together with neck resistance, but it is not a per-spine reanalysis, not biological validation, not a prevalence estimate, and not a controversy resolution. A true empirical test would require paired same-spine R_{neck} , local dendritic load, and transfer definitions from the same preparation.

The current framework remains intentionally limited. The baseline soma is a lumped downstream load, passive morphologies are procedural rather than cell-specific reconstructions, active conductances are generic stress tests rather than calibrated densities, and uncertainty ensembles are deterministic sensitivity designs rather than posterior biological distributions. SPINE focuses on electrical behavior and does not model biochemical calcium reaction-diffusion, receptor trafficking, signaling cascades, or structural plasticity.

The most valuable next studies would combine voltage imaging, morphology measurements, and electrophysiology in preparations where neck-radius uncertainty can be propagated explicitly. Modeling extensions should prioritize calibrated cell-type active conductances, morphology-diverse posterior inference, analysis-defined high- ρ_L uncertainty coverage, and fuller native cross-simulator checks. Accordingly, the strongest supported claim is that the divider is expected and the residuals are informative.

The load-normalized spine-neck ratio is therefore a compact coordinate for the classical local voltage divider, not a universal electrical phenotype. Its most defensible use is to state the expected low-frequency head-to-dendrite transfer and then measure the residual from that expectation. In SPINE, those residuals show when transient conductance dynamics, impedance, synaptic drive, active state, morphology, downstream filtering, and measurement uncertainty make the scalar descriptor incomplete.

3 Online Methods

3.1 Reference circuit and load-normalized ratio

SPINE was implemented in Python using project-local equations, matrix assembly, solvers, metrics, and validation scripts. General scientific Python libraries are used for numerical linear algebra and analysis, but the runtime implementation does not rely on a prebuilt neural simulation engine. Baseline, plausibility-revised, active-extension, and exploratory configurations are separated so that extensions are not silently folded into the baseline reference track.

The baseline reference model contains a spine head h , a parent dendrite compartment d , and a soma/load compartment s . For a cylindrical spine neck with length L_n , radius r_n , and intracellular resistivity ρ_i ,

$$R_{\text{neck}} = \frac{\rho_i L_n}{\pi r_n^2}, \quad (3)$$

with geometry converted to centimeters before resistance is computed. The baseline target neck geometries are listed in Table 1. The head area is approximated as $4\pi r_h^2$ and sets head capacitance

and leak; dendrite and soma capacitances and leaks are lumped. The soma is therefore a compact downstream load rather than a reconstructed full neuron.

The dendrite-soma input resistance $R_{\text{in},d}$ is measured after removing the stimulated spine and neck edge. A small current perturbation is applied to the dendrite-soma subcircuit, and $\Delta V_d/\Delta I_d$ is computed from the steady-state passive linear system with a perturbation simulation check. We then define the load-normalized spine-neck ratio

$$\rho_L = \frac{R_{\text{neck}}}{R_{\text{in},d}}. \quad (4)$$

The code and some provenance tables retain SMI as a legacy shorthand for ρ_L . This spine-omitted definition avoids normalizing a neck by a load that already contains the same neck. One-port reconstructions showed that attached-versus-omitted $R_{\text{in},d}$ differences were numerically negligible for the compact passive-head model family, but the omitted definition remains conceptually cleaner.

3.2 Analytic divider and residual target

The analytic local reference is the DC small-signal voltage divider:

$$\Gamma_{\text{div}} = \frac{R_{\text{in},d}}{R_{\text{neck}} + R_{\text{in},d}} = \frac{1}{1 + \rho_L}. \quad (5)$$

This equation is treated as classical circuit logic, not as a novel result. The manuscript's main diagnostic quantity is the residual between the simulated peak local transfer and this divider expectation:

$$\Delta_{\text{div}} = \Gamma_{h \rightarrow d} - \Gamma_{\text{div}}. \quad (6)$$

Here $\Gamma_{h \rightarrow d} = A_d/A_h$ is measured from transient conductance-based simulations over the analysis-defined metric window. In the primary residual table, A_h , A_d , and A_s are baseline-subtracted maxima measured over the same 50 ms post-event window but not required to occur at the same time. Thus $\Gamma_{h \rightarrow d}$ is a separate-peak transient summary, whereas Γ_{div} is a DC low-frequency reference. Nonzero residuals are expected whenever peak timing, capacitance, changing synaptic driving force, morphology, impedance, active state, or downstream coupling matters. Matched-time and area-under-curve ratios are therefore reported only as targeted controls, not as replacements for the manuscript's primary peak residual.

3.3 Passive equations and numerical solution

Passive voltages obey conductance-balance equations for leak, synaptic, head-dendrite, and dendrite-soma terms:

$$C_h \frac{dV_h}{dt} = -g_{L,h}(V_h - E_L) - g_{\text{syn}}(t)(V_h - E_{\text{syn}}) + g_{hd}(V_d - V_h), \quad (7)$$

$$C_d \frac{dV_d}{dt} = -g_{L,d}(V_d - E_L) + g_{hd}(V_h - V_d) + g_{ds}(V_s - V_d), \quad (8)$$

$$C_s \frac{dV_s}{dt} = -g_{L,s}(V_s - E_L) + g_{ds}(V_d - V_s). \quad (9)$$

Letting $\mathbf{V} = (V_h, V_d, V_s)^T$ and $\mathbf{C} = \text{diag}(C_h, C_d, C_s)$ gives

$$\mathbf{C} \frac{d\mathbf{V}}{dt} = -\mathbf{G}(t)\mathbf{V} + \mathbf{b}(t), \quad (10)$$

where

$$\mathbf{G}(t) = \begin{bmatrix} g_{L,h} + g_{\text{syn}}(t) + g_{hd} & -g_{hd} & 0 \\ -g_{hd} & g_{L,d} + g_{hd} + g_{ds} & -g_{ds} \\ 0 & -g_{ds} & g_{L,s} + g_{ds} \end{bmatrix}. \quad (11)$$

Backward Euler advances the passive state by solving

$$\left(\frac{\mathbf{C}}{\Delta t} + \mathbf{G}_{n+1} \right) \mathbf{V}_{n+1} = \frac{\mathbf{C}}{\Delta t} \mathbf{V}_n + \mathbf{b}_{n+1}. \quad (12)$$

Crank-Nicolson comparisons are used as internal numerical self-consistency checks, not as independent external validation.

The synapse is conductance based:

$$I_{\text{syn},h}(t, V_h) = g_{\text{syn}}(t)(V_h - E_{\text{syn}}), \quad (13)$$

where $g_{\text{syn}}(t)$ is a normalized double exponential with peak conductance g_{max} . Peak depolarizations A_h , A_d , and A_s are measured after subtracting the pre-event baseline over an analysis-defined 50 ms post-event window. Attenuation ratios are $\Gamma_{h \rightarrow d} = A_d/A_h$ and $\Gamma_{h \rightarrow s} = A_s/A_h$. $\Gamma_{h \rightarrow d}$ measures local transfer across the neck, $\Gamma_{h \rightarrow s}$ combines local transfer with downstream dendrite-to-soma filtering, and A_h measures local voltage excursion.

3.4 Challenge suites and descriptor comparisons

Passive morphology analyses generalize the reference circuit to compartment trees assembled from nodes and axial edges. Frequency-domain analyses solve the linearized passive system to estimate input impedance, transfer impedance, gain, phase, and dynamic ratio variants. Active and NMDA mechanisms are disabled in the baseline reference configuration and enabled only in active-extension protocols. These active conductances are generic stress tests using literature-traceable current forms [18–24], not calibrated cell-type-specific channel densities.

Descriptor comparisons were performed on existing derived rows. We compared raw ρ_L , the analytic divider transform Γ_{div} , component resistances R_{neck} and $R_{\text{in},d}$, log-component models, impedance and dynamic descriptors, transfer gain, and synaptic conductance scale where available. These comparisons were interpreted by target and descriptor family. Exact winner labels were avoided when near-tied or correlated predictors made the ranking unstable.

Designed counterexamples tested logical sufficiency. Equal- ρ_L cases ask whether equal ratio implies equivalent voltage responses; iso-neck-resistance cases ask whether equal R_{neck} can change under different loads; iso-transfer cases ask whether similar transfer can arise from different ratios. These experiments were designed to expose limitations as well as successes.

3.5 Deterministic uncertainty and reporting language

Uncertainty analyses used deterministic Latin-hypercube ensembles and progressive sample sizes of $N=96, 192, 384$, and 768. The global uncertainty seed was 202605; progressive ensembles used $202605 + 5100 + N$. Radius-only uncertainty was analyzed separately because $R_{\text{neck}} \propto 1/r_n^2$, using 25 deterministic offsets spanning ± 2 SD with $\text{SD} = 0.018 \mu\text{m}$ and clipping measured radius to $0.06\text{--}0.22 \mu\text{m}$. Heuristic ratio classes were low < 0.25 , intermediate 0.25 to < 0.75 , and high ≥ 0.75 .

All uncertainty fractions are reported as deterministic design fractions over sampled parameter combinations. They are not biological rates and are not treated as inferential population intervals. The $N=768$ ensemble contained 757 low- ρ_L rows, 11 intermediate- ρ_L rows, and zero high- ρ_L rows, so high-ratio uncertainty claims are explicitly scoped to reference, designed challenge rows, and the labeled targeted high- ρ_L extension rather than to the main uncertainty ensemble.

3.6 External summary-anchor decision gate

External summary rows were taken only from the locked Phase 1 summary dataset generated by a Codex-only acquisition gate. No manual transcription, figure digitization, new source acquisition, per-spine reconstruction, or hidden load imputation was used. Row eligibility required a machine-recovered summary or model value for both R_{neck} and a local dendritic load. This admitted three summary-usable rows: Harnett model/summary, Popovic measured summary, and Popovic model-fit [6, 13]. Kwon and Cornejo remained context/exclusion rows because no accepted paired R_{neck} /load coordinate was available [7, 9].

For each eligible row, source-native coordinates were computed as $\rho_L = R_{\text{neck}}/\text{load}$ and $\Gamma_{\text{div}} = 1/(1 + \rho_L)$. Load sensitivity used source-native loads, fixed/recovered common loads of 144.487, 125, 275, and 564 M Ω , and a deterministic diagnostic grid from 75 to 666.667 M Ω ; these load choices are stress-test references, not a biological distribution. Classes used the same heuristic thresholds as the deterministic uncertainty analysis: low < 0.25 , intermediate 0.25 to < 0.75 , and high ≥ 0.75 . For a fixed R_{neck} , the high-threshold load is $R_{\text{neck}}/0.75$ and the low-threshold load is $R_{\text{neck}}/0.25$. Source-reported uncertainty was propagated only for the Popovic measured row, using the locked 27 ± 6 M Ω neck resistance and 275 ± 27 M Ω load envelope; no uncertainty was invented for Harnett or the Popovic model-fit row. External transfer ratios were retained only as proxy compatibility checks and not SPINE residuals.

3.7 Computational credibility checks

Numerical credibility was assessed at three levels. First, internal checks covered units, current signs, synaptic normalization, limiting cases, timestep refinement, sparse/dense passive assembly, impedance verification, and Backward Euler versus Crank-Nicolson peak differences. Second, an independent direct-matrix benchmark separately assembled the baseline passive three-compartment equations and compared against existing SPINE trace outputs without calling the production passive simulation function as its benchmark engine. Third, DC analytic checks tested the two-node dendrite-soma input resistance, divider formula, limiting cases, and attached-versus-omitted one-port behavior.

NEURON and Brian2 importability were checked in the default project runtime. NEURON was unavailable and is not claimed. Brian2 was unavailable in that runtime but was installed in a project-local optional validation environment, where it was used with explicit units, a Brian2 clock/state container, and an independent SI backward-Euler update. The Brian2-hosted scoped checks covered the three passive baseline cases, one cylindrical distributed-neck passive case, and one AMPA/NMDA synapse-only case. Native Brian2 state updaters were attempted but not retained because explicit/RK updates were unstable for the stiff compact head-neck system and the coupled symbolic linear updater was not practical in this local environment. The executed checks therefore support representative computational credibility within scope; they do not validate all rows, restrained active channels, or biological calibration.

Environment, optional Brian2 versions, solver method, timestep, analysis window, deterministic seeds, release tag, and DOI are summarized in Supplementary Table S7.

3.8 Traceability

All manuscript-facing values are rounded for readability while exact values remain in machine-readable CSV outputs and ledgers. The claim-to-source ledger links major quantitative statements to source reports and CSV rows, following model-sharing and FAIR-data principles [25, 26]. Figure

and table manifests record provenance for publication assets. Exploratory perturbation scenarios remain supplemental stress tests and are not used to support primary biological mechanism claims.

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Author contributions: Gregory Pierpoint: Methodology; Software; Formal analysis; Investigation; Validation; Visualization; Data curation; Writing – original draft; Writing – review & editing. Alberto Musto: Conceptualization; Methodology; Supervision; Project administration; Writing – review & editing; Scientific interpretation; Final manuscript approval.

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Data Availability

This study uses mathematical and computational modeling and generated no new human participant or animal data. Manuscript-facing derived data, validation outputs, figure/table source data, and provenance ledgers are included in the SPINE release package. Code, release metadata, and manuscript-facing derived data are archived at <https://github.com/blakepi/spine-divider-residuals> and <https://doi.org/10.5281/zenodo.20672333>. The release tag is **v1.0.0-submission**.

Manuscript claim provenance is recorded in `manuscript/CLAIM_TO_SOURCE_LEDGER.csv`. Figure and table provenance are recorded in `manuscript/FIGURE_SOURCE_MANIFEST.csv` and `manuscript/TABLE_SOURCE_MANIFEST.csv`. These ledgers are traceability aids and do not replace the manuscript’s scoped claims.

Code Availability

The SPINE Python code, configuration files, analysis scripts, tests, validation scripts, environment instructions, release manifests, and manuscript-facing derived data are available at <https://github.com/blakepi/spine-divider-residuals> and archived at <https://doi.org/10.5281/zenodo.20672333> [27]. The release tag is **v1.0.0-submission**. Code is released under the MIT License. Derived manuscript-facing data, figures, tables, and documentation are released under CC BY 4.0 unless otherwise stated.

References

- [1] Wilfrid Rall. Theoretical significance of dendritic trees for neuronal input-output relations. In *Neural Theory and Modeling*, pages 73–97. Stanford University Press, Stanford, CA, 1964. doi: 10.7551/mitpress/6743.003.0015.
- [2] Idan Segev and Wilfrid Rall. Computational study of an excitable dendritic spine. *Journal of Neurophysiology*, 60(2):499–523, 1988. doi: 10.1152/jn.1988.60.2.499.

- [3] David Tsay and Rafael Yuste. On the electrical function of dendritic spines. *Trends in Neurosciences*, 27(2):77–83, 2004. doi: 10.1016/j.tins.2003.11.008.
- [4] Rafael Yuste. Electrical compartmentalization in dendritic spines. *Annual Review of Neuroscience*, 36:429–449, 2013. doi: 10.1146/annurev-neuro-062111-150455.
- [5] Roberto Araya, Jiang Jiang, Kenneth B. Eisenthal, and Rafael Yuste. The spine neck filters membrane potentials. *Proceedings of the National Academy of Sciences of the United States of America*, 103(47):17961–17966, 2006. doi: 10.1073/pnas.0608755103.
- [6] Mark T. Harnett, Judit K. Makara, Nelson Spruston, William L. Kath, and Jeffrey C. Magee. Synaptic amplification by dendritic spines enhances input cooperativity. *Nature*, 491(7425):599–602, 2012. doi: 10.1038/nature11554.
- [7] Taekyung Kwon, Miki Sakamoto, Darcy S. Peterka, and Rafael Yuste. Attenuation of synaptic potentials in dendritic spines. *Cell Reports*, 20(5):1100–1110, 2017. doi: 10.1016/j.celrep.2017.07.012.
- [8] L. Beaulieu-Laroche and Mark T. Harnett. Dendritic spines prevent synaptic voltage clamp. *Neuron*, 97(1):75–82.e3, 2018. doi: 10.1016/j.neuron.2017.11.016.
- [9] Victor Hugo Cornejo, Netanel Ofer, and Rafael Yuste. Voltage compartmentalization in dendritic spines in vivo. *Science*, 375(6576):82–86, 2022. doi: 10.1126/science.abg0501.
- [10] Jeffrey C. Magee. Dendritic integration of excitatory synaptic input. *Nature Reviews Neuroscience*, 1(3):181–190, 2000. doi: 10.1038/35044552.
- [11] Michael London and Michael Hausser. Dendritic computation. *Annual Review of Neuroscience*, 28:503–532, 2005. doi: 10.1146/annurev.neuro.28.061604.135703.
- [12] Greg J. Stuart and Nelson Spruston. Dendritic integration: 60 years of progress. *Nature Neuroscience*, 18(12):1713–1721, 2015. doi: 10.1038/nn.4157.
- [13] Marko A. Popovic, Nicholas Carnevale, Balázs Rozsa, and Dejan Zecevic. Electrical behaviour of dendritic spines as revealed by voltage imaging. *Nature Communications*, 6:8436, 2015. doi: 10.1038/ncomms9436.
- [14] Dejan Zecevic. Electrical properties of dendritic spines. *Biophysical Journal*, 122(22):4303–4315, 2023. doi: 10.1016/j.bpj.2023.10.008.
- [15] Jeffrey C. Magee and Erik P. Cook. Somatic epsp amplitude is independent of synapse location in hippocampal pyramidal neurons. *Nature Neuroscience*, 3(9):895–903, 2000. doi: 10.1038/78800.
- [16] Jan Tønnesen, Gergely Katona, Balázs Rózsa, and U. Valentin Nägerl. Spine neck plasticity regulates compartmentalization of synapses. *Nature Neuroscience*, 17(5):678–685, 2014. doi: 10.1038/nn.3682.
- [17] Jan Tønnesen and U. Valentin Nägerl. Dendritic spines as tunable regulators of synaptic signals. *Frontiers in Psychiatry*, 7:101, 2016. doi: 10.3389/fpsy.2016.00101.
- [18] Alan L. Hodgkin and Andrew F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4):500–544, 1952. doi: 10.1113/jphysiol.1952.sp004764.

- [19] Mark L. Mayer, Gary L. Westbrook, and Peter B. Guthrie. Voltage-dependent block by mg^{2+} of nmda responses in spinal cord neurones. *Nature*, 309:261–263, 1984. doi: 10.1038/309261a0.
- [20] Craig E. Jahr and Charles F. Stevens. A quantitative description of nmda receptor-channel kinetic behavior. *The Journal of Neuroscience*, 10(6):1830–1837, 1990. doi: 10.1523/JNEUROSCI.10-06-01830.1990.
- [21] Jeffrey C. Magee. Dendritic hyperpolarization-activated currents modify the integrative properties of hippocampal ca1 pyramidal neurons. *The Journal of Neuroscience*, 18(19):7613–7624, 1998. doi: 10.1523/JNEUROSCI.18-19-07613.1998.
- [22] Jeffrey C. Magee. Dendritic ih normalizes temporal summation in hippocampal ca1 neurons. *Nature Neuroscience*, 2(6):508–514, 1999. doi: 10.1038/9158.
- [23] John A. Connor and Charles F. Stevens. Prediction of repetitive firing behaviour from voltage clamp data on an isolated neurone soma. *The Journal of Physiology*, 213(1):31–53, 1971. doi: 10.1113/jphysiol.1971.sp009366.
- [24] John R. Huguenard and David A. McCormick. Simulation of the currents involved in rhythmic oscillations in thalamic relay neurons. *Journal of Neurophysiology*, 68(4):1373–1383, 1992. doi: 10.1152/jn.1992.68.4.1373.
- [25] Michael L. Hines, Thomas Morse, Michele Migliore, Nicholas T. Carnevale, and Gordon M. Shepherd. Modeldb: A database to support computational neuroscience. *Journal of Computational Neuroscience*, 17(1):7–11, 2004. doi: 10.1023/B:JCNS.0000023869.22017.2e.
- [26] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J. G. Gray, Paul Groth, Carole Goble, Jeffrey S. Grethe, Jaap Heringa, Peter A. C. 't Hoen, Rob Hooft, Tobias Kuhn, Ruben Kok, Joost Kok, Scott J. Lusher, Maryann E. Martone, Albert Mons, Abel L. Packer, Bengt Persson, Philippe Rocca-Serra, Marco Roos, Rene van Schaik, Susanna-Assunta Sansone, Erik Schultes, Thierry Sengstag, Ted Slater, George Strawn, Morris A. Swertz, Mark Thompson, Johan van der Lei, Erik van Mulligen, Jan Velterop, Andra Waagmeester, Peter Wittenburg, Katherine Wolstencroft, Jun Zhao, and Barend Mons. The fair guiding principles for scientific data management and stewardship. *Scientific Data*, 3(1):160018, 2016. doi: 10.1038/sdata.2016.18.
- [27] Gregory Blake Pierpoint and Alberto E. Musto. Spine: reproducible compartmental-model analyses of residual limits of a load-normalized spine-neck divider coordinate, 2026. URL <https://doi.org/10.5281/zenodo.20672333>.